

IntrusionX Second Edition

PhishShield AI

Preventing credential theft before it happens



DIGITAL-DETECTIVES



Cybersecurity + AI





PROPOSED SOLUTION:

- PhishShield AI is a **browser extension** that detects phishing websites in real time
- Uses a **hybrid approach**:
- Machine Learning (pattern detection)
- Behavioral analysis (page activity)
- Risk scoring (decision system)

How it solves the problem

- Detects phishing **before user enters credentials**
- Avoids false positives using **context-aware scoring**
- Provides **Explainable AI (why flagged)**

Innovation & Uniqueness

- Hybrid AI system (ML + rules + behavior)
- Real-time detection inside browser
- Transparent decision-making (user trust)

TECHNICAL APPROACH

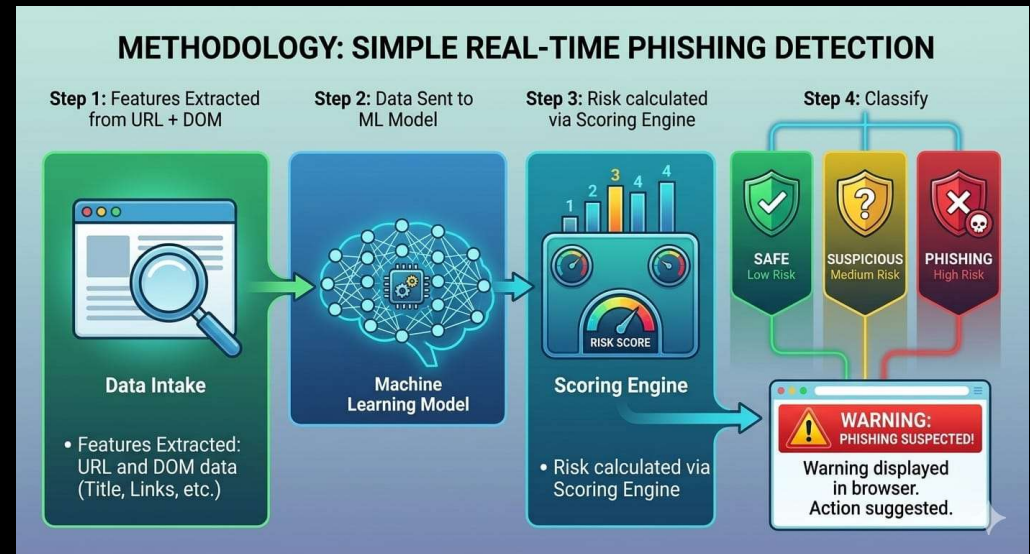


Technologies Used

- Frontend: HTML, CSS, JavaScript
- Extension: Chrome Extension APIs
- Backend: Python (Flask)
- ML Model: Random Forest (Scikit-learn)

Methodology

- Extract features from URL + DOM
- Send data to ML model
- Calculate risk using scoring engine
- Classify:
 - Safe / Suspicious / Phishing
- Display real-time warning popup



FEASIBILITY AND VIABILITY



1. Feasibility

Lightweight Chrome extension → easy deployment
Works in real-time without heavy resources
ML model already trained and integrated

2. Challenges

False positives on login pages
Limited feature extraction
Dynamic website behavior

3. Solutions

Introduced weighted risk scoring
Added DOM-based feature extraction
Hybrid ML + rule-based system

IMPACT AND BENEFITS



Impact

Prevents credential theft in real time
Protects users from phishing attacks
Improves user awareness with explanations

Benefits

- ✓ High accuracy with reduced false positives
- ✓ Explainable AI builds trust
- ✓ Scalable to millions of users
- ✓ Applicable for students, enterprises, browsers

RESEARCH AND REFERENCES



- **UCI Machine Learning Repository**
Phishing Websites Dataset (used for model training)
- **PhishTank (Open Phishing Database)**
Real-world phishing URLs for validation and testing
- **Scikit-learn Documentation**
Used for implementing the Random Forest ML model
- **Google Safe Browsing (Concept Reference)**
Inspiration for real-time phishing detection approach
- **OWASP Phishing Prevention Guidelines**
Security best practices for detecting phishing attacks

“System designed using real-world cybersecurity practices and datasets.”